

How Accurate Are Learned World Models? Measuring DreamerV3’s Imagination Fidelity on Snake

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Abstract

DreamerV3 learns a world model from experience and trains its policy entirely inside that learned model—it learns by dreaming rather than by doing. This approach works across dozens of domains without changing a single hyperparameter, but how accurate are those dreams? This work trains DreamerV3 on Snake—a game with simple, deterministic dynamics that make imagination errors precisely measurable—and extracts imagined trajectories at 20 checkpoints across one million training steps. Each imagined frame is compared against the real game state using both pixel-level metrics (MSE, SSIM) and semantic metrics that capture whether the model correctly places the snake head, body segments, and food. The results show that imagination accuracy improves significantly over training (Spearman $\rho = -0.85$ for head position error, $p < 0.001$), and that imagination quality correlates strongly with game performance ($\rho = -0.84$ between head error and eval return). Head position accuracy emerges as the single most predictive metric, outperforming both pixel-level measures and other semantic metrics. Training is ongoing, with additional runs planned to test whether these patterns hold across different hyperparameter configurations.

Introduction

Model-based reinforcement learning agents learn a model of the environment and use it for planning, rather than learning a policy directly from interactions (Sutton and Barto 2018). The Dreamer family of algorithms takes this approach to its logical extreme: the agent trains its policy entirely inside the learned world model, never touching the real environment during policy optimization (Hafner et al. 2020, 2021, 2025). DreamerV3 represents the current state of the art, mastering Atari games, continuous control, and even Minecraft—all with the same fixed hyperparameters (Hafner et al. 2025).

Most research on DreamerV3 focuses on end-task performance: how high the score is, how fast learning converges. What gets less attention is the world model itself. The agent trains inside an imagined version of the environment. So how faithful is that imagination to reality? If the world model predicts the wrong outcome for an action, the agent optimizes for a scenario that does not exist. Understanding when and how imagination diverges from reality could

reveal why model-based RL works as well as it does—and where it might break down.

The central question is: **Does DreamerV3 need accurate imagination to learn effective policies, or can it succeed even when its world model is wrong?** This decomposes into three testable hypotheses:

- H1:** Imagination accuracy improves as training progresses—the world model learns to dream better over time.
- H2:** Better imagination leads to better play—imagination accuracy correlates with game performance.
- H3:** Semantic accuracy matters more than pixel accuracy—knowing where game elements are is more important than producing sharp images.

This work investigates these hypotheses by training DreamerV3 on Snake, a game with simple, deterministic dynamics. Snake is an ideal testbed because every transition except food placement is fully predictable: the snake moves one cell per step, the game ends on collision, and the state is fully visible. There is no visual clutter, no stochastic physics, no ambiguity. When the world model imagines a trajectory, each frame can be compared against the actual game state with perfect precision.

The contributions of this work are: (1) a systematic pipeline for extracting and evaluating DreamerV3’s imagined trajectories against ground-truth game states, (2) evidence from 20 training checkpoints that imagination accuracy improves significantly over training and correlates with game performance, and (3) a finding that head position accuracy is the single strongest predictor of agent performance—stronger than any pixel-level metric.

This paper presents interim results from one completed training run. Additional training configurations are planned for the final paper to test whether these patterns generalize.

Background

World Models in Reinforcement Learning

The idea of learning a model of the environment for planning has a long history in RL (Sutton and Barto 2018). Ha and Schmidhuber (2018) showed that an agent could learn a compressed representation of its environment using a variational autoencoder and a recurrent network, then train a policy entirely within that learned model. This demonstrated

that learning from imagined experience could produce effective behavior without additional environment interaction.

Hafner et al. (2019) formalized this approach with the Recurrent State-Space Model (RSSM), which maintains both a deterministic recurrent state (capturing persistent temporal information) and a stochastic latent state (modeling uncertainty). The RSSM became the backbone of all subsequent Dreamer agents.

The Dreamer Family

DreamerV1 introduced actor-critic training entirely within imagined trajectories generated by the RSSM (Hafner et al. 2020). DreamerV2 improved on this with discrete latent representations, replacing Gaussian latents with categorical variables that produced sharper predictions and achieved human-level Atari performance (Hafner et al. 2021). DreamerV3 further refined the approach with symlog predictions for handling varying reward scales and a fixed hyperparameter set that works across radically different domains (Hafner et al. 2025). It was the first single algorithm to collect diamonds in Minecraft from scratch.

Related Work on World Model Analysis

Several works have extended or modified the Dreamer architecture. Deng, Jang, and Ahn (2022) removed the reconstruction objective, training the world model with prototypical representations instead. Yuan et al. (2023) extended the RSSM to handle irregularly sampled data. Sekar et al. (2020) used the world model for self-supervised exploration without task rewards. Lin et al. (2024) incorporated language into the world model for instruction-following.

However, these works focus on improving the world model or the policy—not on measuring how accurate the imagination actually is. Walker et al. (2023) investigated model-based learning’s role in exploration and transfer, and Sullivan et al. (2023) analyzed whether DreamerV3’s design choices transfer to model-free methods. A direct, systematic measurement of imagination fidelity—comparing dreamed trajectories frame-by-frame against ground truth—remains largely unexplored. This gap matters because understanding what the world model gets right and wrong could inform architectural improvements, reveal what representations are actually learned, and explain why model-based RL generalizes as well as it does.

Snake as an MDP

Snake is formalized as a Markov Decision Process ($\mathcal{S}, \mathcal{A}, P, R, \gamma$). The state space consists of a 10×10 grid containing the snake head, body segments, and food, rendered as 64×64 RGB images. The action space contains four discrete actions (up, down, left, right), with 180-degree reversals blocked. Transitions are deterministic given state and action: the snake moves one cell per step, grows on eating food, and dies on collision. The reward is $+1$ for food, -1 for collision, and -0.002 per step to discourage idle behavior. The discount $\gamma = 0.999$ encourages long-horizon play.

The simplicity and determinism of Snake’s dynamics make it uniquely suited for imagination analysis. Every

frame in an imagined trajectory can be precisely validated against the true game state, enabling measurement of exactly what the world model learns to predict and where it fails.

Methods

Environment and Training

A Gymnasium-compatible Snake environment was implemented with a 10×10 grid, 64×64 RGB observations, and distinct colors for the snake head (bright green), body (dark green), food (red), and background (black). These colors enable automated state extraction from both real and imagined frames.

DreamerV3 was trained using a publicly available PyTorch implementation with the snake-specific configuration shown in Table 1. The agent was trained for approximately 1.1 million environment steps, producing 21 checkpoints at roughly 50,000-step intervals. The training run stored 252 evaluation episodes (10 episodes per evaluation round) alongside training metrics.

Hyperparameter	Value
Grid size	10×10
Observation size	$64 \times 64 \times 3$
Max steps per episode	500
Training steps	$\sim 1.1\text{M}$
Parallel environments	4
Batch size	16
Batch length	64
Train ratio	256
Model learning rate	1×10^{-4}
Actor/Critic LR	3×10^{-5}
Discount (γ)	0.999
RSSM deterministic dim	512
RSSM stochastic dim	32×32 (discrete)
Imagination horizon	15
Step penalty	-0.002

Table 1: DreamerV3 training hyperparameters for the Snake environment.

Imagination Extraction

Imagined trajectories are extracted from each checkpoint using the following procedure:

1. Load the world model weights from the checkpoint, stripping the `torch.compile` prefix used during training.
2. Select 20 evaluation episodes from the stored recordings.
3. For each episode, feed the first 5 frames through the encoder and RSSM posterior (the “context” phase), establishing the latent state.
4. From the final posterior state, roll the RSSM forward using the real action sequence—but without any further observations—for up to 55 additional steps (the “imagination” phase).

5. Decode each imagined latent state through the CNN decoder to produce RGB frames.

This mirrors the `video_pred` method already present in DreamerV3’s world model class, which uses the same encode→observe→imagine→decode pipeline. The 5-frame context length matches the convention used in DreamerV3’s built-in video prediction logging.

The result is paired sequences of real and imagined frames starting from the same initial state and following the same action sequence. The only difference between them is the world model’s prediction accuracy.

Metrics

Imagination accuracy is evaluated at two levels:

Frame-level metrics capture overall visual fidelity:

- **MSE:** Mean Squared Error between real and imagined frames (float32, 0–1 range). Lower is better.
- **SSIM:** Structural Similarity Index (Wang et al. 2004) computed on uint8 frames. Higher is better.

Semantic metrics capture game-state accuracy by extracting the head, body, and food positions from both real and imagined frames using a color-matching state extractor:

- **Head error:** Euclidean distance (in grid cells) between real and imagined head positions. A value of 0 means the model places the head in the correct cell.
- **Body accuracy:** Fraction of real body segments that are present in the imagined frame (0–1).
- **Food correct:** Whether the imagined frame places food in the correct cell (binary).

The state extractor works by sampling the center pixel of each 6×6 grid cell and matching it to the nearest reference color (head, body, food, or background) using L2 distance in RGB space. A distance threshold of 100 handles the blurry, imperfect outputs of the decoder while maintaining reliable classification.

Correlation Analysis

To test the three hypotheses, Spearman rank correlations are computed:

- **H1:** Correlation between training step and each imagination metric across checkpoints.
- **H2:** Correlation between each imagination metric and the agent’s evaluation return (average reward per episode) at the corresponding checkpoint.
- **H3:** Comparison of correlation strengths between pixel-level and semantic metrics using Fisher’s z -test on the highest-correlation metric from each category.

Statistical significance is assessed at $\alpha = 0.05$ throughout.

Experiments

Training Progress

The agent was trained for approximately 1.1 million environment steps. Evaluation return increased from near zero

at 55K steps to approximately 25.7 at 1.05M steps. The agent learned to navigate the grid, collect food, and avoid collisions, with average episode lengths stabilizing around 400–475 steps in later checkpoints.

Early in training (55K–155K steps), the agent achieved low scores (0.2–0.4 mean return) and short survival. By 505K steps, mean return reached 8.2, and by the final checkpoint (1.05M steps), it reached 25.7—indicating the agent consistently collects 25+ pieces of food per episode.

Imagination Extraction

Imagined trajectories were extracted from all 21 checkpoints (55K through 1.1M steps), processing 20 evaluation episodes per checkpoint with a 5-frame context and up to 55 imagination steps. This produced 916 frame pairs per checkpoint (420 episodes total, 19,236 total frame comparisons).

Hardware

All training and analysis was performed on a single workstation with an NVIDIA RTX 5070 Ti (16 GB VRAM), AMD Ryzen 9 9900X, and 64 GB RAM, running PyTorch 2.7.0 with CUDA 12.8 on Ubuntu (WSL2). The full analysis pipeline (imagination extraction + metrics + correlation analysis) completed in under 10 minutes for all 21 checkpoints.

Results

H1: Imagination Accuracy Improves Over Training

Table 2 shows Spearman rank correlations between training step and each imagination metric across 20 matched checkpoints. Four of five metrics show statistically significant improvement over training.

Metric	ρ	p -value	Trend
MSE ($\downarrow$)	−0.549	0.012*	Improving
SSIM ($\uparrow$)	+0.654	0.002**	Improving
Head error ($\downarrow$)	−0.850	< 0.001***	Improving
Body accuracy ($\uparrow$)	+0.355	0.125	Improving (n.s.)
Food correct ($\uparrow$)	+0.583	0.007**	Improving

Table 2: H1: Spearman correlations between training step and imagination metrics. Arrows indicate the direction of improvement. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Head position error shows the strongest improvement ($\rho = -0.850$, $p < 0.001$), dropping from a mean of 0.268 grid cells at 55K steps to 0.023 at the final checkpoint. Body accuracy is the only metric that does not reach significance, likely because it starts high (0.948) and has limited room to improve.

The world model’s imagination improves substantially and systematically as training progresses. This supports H1.

Metric	Spearman ρ	p	Pearson r	p
MSE	-0.570	0.009**	-0.442	0.039
SSIM	+0.630	0.003**	+0.444	0.050
Head error	-0.843	< 0.001***	-0.661	0.001***
Body accuracy	+0.352	0.128	+0.347	0.134
Food correct	+0.597	0.005**	+0.457	0.042*

Table 3: H2: Correlations between imagination metrics and evaluation return.

H2: Imagination Correlates with Performance

Table 3 shows correlations between imagination metrics and the agent’s evaluation return.

Head error again shows the strongest relationship: $\rho = -0.843$ ($p < 0.001$). As the world model gets better at predicting where the snake’s head will be, the agent’s game performance improves. MSE, SSIM, and food accuracy also show significant Spearman correlations. Body accuracy does not reach significance, consistent with its ceiling effect.

These results support H2: better imagination is associated with better play. Head position accuracy is the strongest driver of this relationship.

H3: Semantic vs. Pixel Metrics

Table 4 compares the predictive strength of pixel-level and semantic metrics.

Type	Metric	$ \rho $	p -value
Pixel	MSE	0.570	0.009
	SSIM	0.630	0.003
Semantic	Head error	0.843	< 0.001
	Body accuracy	0.352	0.128
	Food correct	0.597	0.005
Average $ \rho $, pixel		0.600	
Average $ \rho $, semantic		0.597	

Table 4: H3: Comparison of pixel-level vs. semantic metric correlations with evaluation return. Fisher’s z -test comparing the best metric from each category (head error vs. SSIM): $z = 1.42$, $p = 0.154$.

The picture for H3 is nuanced. The single best predictor of performance is a semantic metric—head position error ($|\rho| = 0.843$)—which substantially outperforms the best pixel metric (SSIM, $|\rho| = 0.630$). However, the *average* across all semantic metrics (0.597) is nearly identical to the pixel average (0.600), because body accuracy is weak. Fisher’s z -test comparing head error to SSIM yields $z = 1.42$, $p = 0.154$ —not statistically significant with the current sample of 20 checkpoints.

H3 receives partial support. Head position accuracy is clearly the most informative single metric, but the average semantic advantage does not reach significance. This may reflect that body accuracy is a poor metric in this setting (it starts near ceiling) rather than a genuine weakness of semantic measurement.

Discussion

Head Position as the Key Representation

The standout finding across all three hypotheses is the dominance of head position accuracy. It improves the most over training ($\rho = -0.850$), correlates the most with performance ($\rho = -0.843$), and is the strongest individual predictor of game return.

This makes intuitive sense. In Snake, the head is the locus of all agency: it determines whether food is eaten, whether collisions happen, and where the body will be next. A world model that knows where the head is effectively knows what is about to happen. Body segments follow deterministically from the head’s history, and food collection depends on the head reaching the food cell. Getting the head right may be sufficient for the policy to extract useful gradients, even if other aspects of the image are slightly off.

Why Body Accuracy is a Weak Signal

Body accuracy starts at 0.948 even at the earliest checkpoint and only improves to 0.985 by the end. This ceiling effect limits its statistical power. But it also reflects a property of the representation: a 10×10 grid with a snake of length 3–5 means that most cells are empty, and predicting “background” for most cells yields high body accuracy by default. Future work could report body accuracy only for occupied cells, or weight errors by snake length, to produce a more informative metric.

Imagination Quality as a Diagnostic Tool

These results suggest that imagination fidelity metrics—particularly game-state-level ones—could serve as diagnostic tools for model-based RL more broadly. Instead of only monitoring reward curves, practitioners could track how well the world model predicts key environment features over training. A plateau in imagination accuracy that precedes a performance plateau, for example, would indicate that the bottleneck has shifted from the world model to the policy.

Limitations and Ongoing Work

Several limitations apply to the current results:

Single training run. All results come from one training configuration. It is not yet known whether the same patterns hold with different learning rates, train ratios, or board sizes. Additional training runs (Table 5) are planned for the final paper.

Short imagination horizon. Imagined trajectories are capped at 60 steps. Longer horizons may reveal different degradation patterns or threshold effects.

No collision accuracy. The proposal included collision prediction accuracy as a planned metric. This has not yet been implemented and will be added for the final paper.

Limited episode diversity. All evaluation episodes come from the same set of 252 recorded episodes. Future work will generate fresh episodes from each checkpoint’s policy for a fairer comparison.

Single environment. Snake is deliberately simple. Whether these findings transfer to environments with richer

visual structure or stochastic dynamics remains an open question.

Run	Change from Baseline
Model 3	Longer episodes (2000 steps), fine-tuned
Model 4	Higher learning rate ($3\times$ baseline)
Model 5	Lower learning rate ($\frac{1}{3}\times$ baseline)
Model 6	Higher train ratio (512 vs 256)
Model 7	Larger board (20×20)

Table 5: Planned training configurations for the final paper. Each varies one hyperparameter from the current baseline.

Conclusion

DreamerV3’s world model produces imagined trajectories that improve significantly over training and correlate with game performance. Among five metrics tested, head position accuracy is the standout predictor—it shows the strongest improvement over training ($\rho = -0.85$, $p < 0.001$) and the strongest correlation with the agent’s actual game score ($\rho = -0.84$, $p < 0.001$). This suggests that the world model’s ability to track the snake’s head—the locus of agency in the game—is the most critical learned representation.

These are interim results from a single completed training run over approximately one million steps. The analysis pipeline is operational and will be applied to additional training configurations to test whether these patterns generalize across hyperparameters and environment complexity. The full results, including collision prediction accuracy and multi-configuration comparisons, will be presented in the final paper.

AI Usage Statement

Claude (Anthropic) was used to assist with formatting ideas into prose, structuring this document in AAAI conference format, and implementing the imagination extraction pipeline. All research questions, hypotheses, methods, analysis decisions, and interpretation of results are the author’s own.

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